

Java String Manipulation - Short Notes with Code

1. What is a Mutable String in Java?

In Java, mutable strings are strings whose content can be changed after creation.

Java provides `StringBuffer` and `StringBuilder` classes for mutable strings. The `String` class is immutable.

Example:

```
StringBuilder sb = new StringBuilder("Hello");  
  
sb.append(" World");  
  
System.out.println(sb); // Output: Hello World
```

2. WAP to reverse a String "PWSKILLS" --> "SLLIKSPW"

Code:

```
String str = "PWSKILLS";  
  
StringBuilder sb = new StringBuilder(str);  
  
sb.reverse();  
  
System.out.println(sb); // Output: SLLIKSPW
```

3. WAP to reverse a sentence while preserving word position: "Think Twice" --> "kniht eciwt"

Code:

```
String sentence = "Think Twice";  
  
String[] words = sentence.split(" ");  
  
StringBuilder result = new StringBuilder();  
  
for (String word : words) {  
    result.append(new StringBuilder(word).reverse()).append(" ");  
}  
  
System.out.println(result.toString().trim()); // Output: kniht eciwt
```

4. WAP to sort a String Alphabetically

Code:

```
String str = "PWSKILLS";
```

```
char[] chars = str.toCharArray();
```

```
Arrays.sort(chars);
```

```
String sorted = new String(chars);
```

```
System.out.println(sorted); // Output: ILLPSSSW
```